

All-pairs Chamber screening: quadratic spline

degree 2; fixed knots (0); gamma grid 0--3 by 0.01; alpha = 0.05

Intervened environments: 53
Excluded ir_2-related environments: 5
Unordered pairs evaluated: 1378
Successful fits: 1378
Pairs with gamma_CV > 0: 857
Same sensitivity and higher CV specificity: 252
Same sensitivity, higher CV specificity, all five causes found: 5

Desired pattern: gamma_CV > 0, equal sensitivity = TP/5, and specificity_CV > specificity_OLS over 12 non-causes.

Top 25 pairs with the desired pattern

TP = rejected direct causes; FP = rejected non-causes

Environment 1	Environment 2	gamma	OLS TP	CV TP	OLS FP	CV FP
l_21_mid	t_vis_2_strong	0.4	5	5	3	1
diode_vis_1_mid	osr_angle_1_weak	0.74	5	5	4	2
diode_vis_2_mid	l_21_mid	0.61	5	5	2	1
l_22_mid	v_c_mid	0.06	5	5	5	4
l_21_mid	osr_angle_2_strong	0.2	5	5	6	5
osr_angle_1_strong	v_angle_2_mid	0.33	3	3	7	2
diode_vis_1_mid	v_c_strong	0.75	3	3	5	1
diode_ir_3_strong	pol_2_strong	0.19	3	3	6	2
l_11_mid	t_vis_3_strong	0.98	3	3	6	2
pol_2_mid	t_ir_1_mid	0.52	3	3	6	2
t_vis_1_weak	v_angle_2_mid	0.72	3	3	6	2
red_mid	t_vis_1_mid	0.28	2	2	5	1
osr_angle_1_weak	t_ir_1_weak	0.42	4	4	4	1
t_vis_2_strong	t_vis_2_weak	0.36	4	4	4	1
t_ir_1_weak	v_angle_1_strong	0.32	4	4	5	2
diode_ir_1_mid	t_vis_3_mid	0.14	4	4	6	3
l_21_mid	v_angle_2_mid	0.82	4	4	6	3
diode_ir_1_strong	t_ir_1_mid	0.58	3	3	4	1
diode_vis_2_mid	t_vis_2_mid	0.21	3	3	4	1
diode_ir_3_mid	v_c_strong	0.46	3	3	5	2
diode_ir_3_strong	l_11_mid	0.54	3	3	5	2
l_22_mid	red_mid	0.51	3	3	5	2
osr_c_mid	pol_2_strong	0.28	3	3	5	2
red_mid	t_vis_3_weak	0.65	3	3	5	2
diode_ir_1_mid	diode_ir_3_mid	0.22	3	3	6	3